Which Imagined Concepts Can Be Steered in LLMs?

Anonymous ACL submission

Abstract

Steering vectors are widely used to control and interpret LLMs, yet benchmarks show that some concepts steer reliably while others barely respond — a variance that is reported but rarely explained. In this research we investigate which kinds of concepts are steerable in Llama-3.1-8B and Gemma-2-9B. We use the RFM method to extract steering vectors (or directions) for approximately 600 concepts drawn from 24 WordNet supersense categories, and validate which of them successfully steer the model. We then build a regression model explaining steerability from concept category, concreteness, polysemy, and word frequency, as well as the share of the top eigenvector during extraction (λ_1/trace). The results help distinguish concepts that behave as single directions — "primitive" for the model — from those that are composite or context-dependent, and offer practitioners an extraction-time indication of when single-vector steering can be trusted.

- TODO: the prose paragraph above predates the final results — update: add age-of-acquisition to the predictor list, replace "λ_1/trace" with "spectrum concentration (participation ratio)", and consider adding one sentence each on the depth-window finding and the cross-model replication.
- Headline: 41.5% of the 600 concepts steer on Llama (249/600, V1∪V6) and 37.3% on Gemma (224/600) under the v4 instrument, and the psycholinguistic laws replicate across BOTH models with all four predictors significant: frequency +, concreteness −, age-of-acquisition −, polysemy −. Spectrum concentration (PR) predicts steerability within both models, at model-specific depths. Frequent, early-acquired, monosemous, abstract concepts steer best.
- Candidate sentence (failure mode): when steering fails, it fails informatively — delivery of the category/frame blend, a diffuse discriminative spectrum, or injection outside the concept's depth window; not silence. (1 sentence)
- Candidate sentence (instrument): measured steerability is instrument-relative — one changed frame sentence doubles the rate on both models — but the laws survive instrument and model change. (1 sentence)
- Candidate sentence (depth): concepts are injectable in depth windows ordered by abstractness; one well-chosen layer nearly matches all-layer steering. (1 sentence)

1 Introduction (1.0p)

- Hook: steering/RePE papers throw heterogeneous concept lists at models; some concepts steer at 90%, others at chance. The variance is treated as noise; we argue it is signal. (1 para)
- The implicit assumption of top-1 eigenvector / PCA-1 / difference-of-means steering: the concept is representable as one direction, injectable with one coefficient, at whatever layer(s) the method uses. (half para)
- Existing evidence that not all features are one-dimensional (Engels et al.) — but no one has asked which concepts behave as single directions, or whether this aligns with human semantic structure. (half para)
- Our question: is steerability systematic and predictable from the linguistic properties of the word — and what does a steering failure actually consist of? (half para)
- Contributions (short lead-in + three numbered items, 2–3 sentences each): (1 para)

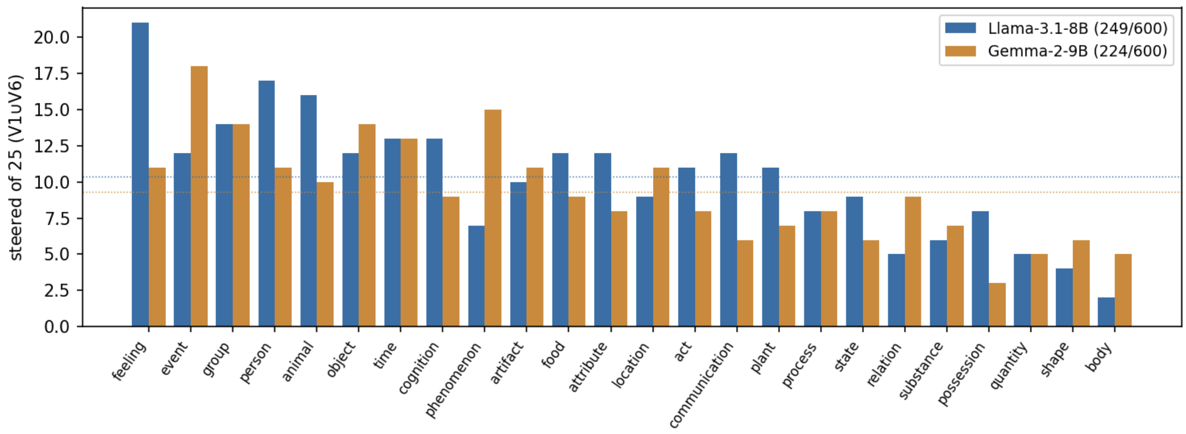

Figure 1: Concepts steered per WordNet supersense (of 25 each; steered = judged as expressing the concept under either evaluation prompt, V1∪V6; all-layer steering, v4 extraction frame): steerability varies systematically by category, and the rates and much of the ordering replicate across the two model families.

- C1: a systematic steerability study — ∼600 lexicalized concepts across 24 WordNet supersenses, annotated with psycholinguistic norms, steered under a controlled protocol on two models — together with a 7-frame instrument study (Table 2) showing that the extraction contrast itself sets measured steerability (2× rate swing at fixed method and model); per-concept results and code released.
- C2: a regression explaining steerability from frequency, concreteness, age-of-acquisition, and polysemy (all significant, signs replicating across Llama and Gemma), with extraction-internal spectral concentration (participation ratio / λ_1-share) as a zero-cost predictor that beats the lexical block.
- C3: a mechanistic account of what failure looks like — parent-dominance of extracted directions, coherence ≠ identity, injection outside the concept's depth window — probed causally (within-category subtraction; single-layer depth sweep).

2 Background and Related Work (1.0p)

- Linear representation hypothesis and steering. Park et al. (2024); RepE (Zou et al., 2023); CAA (Panickssery et al., 2024); ActAdd (Turner et al., 2023); ITI (Li et al., 2023); the RFM benchmark we build on (Beaglehole et al., 2025): kernel ridge + AGOP, top eigenvector as steering vector. Ours: we adopt RFM unchanged and ask which concepts it works for. Second relation: steering rates are instrument-relative — one changed contrast sentence doubles the rate (19.8%→41.5%) at fixed method, model, and judge; method comparisons with hand-tuned per-class prompts inherit this confound. (1 para)
- Multi-dimensional features. Engels et al. (2025): not all features are one-dimensional. Ours: we test the premise on a stratified inventory and tie it to word properties: spectrum concentration (PR / λ_1-share) predicts steering in both models — "effectively one-dimensional" is measurable and graded, and polysemy acts on steerability largely through it. (half para)
- Hierarchical concept geometry. Park et al. (2025): ∼900 WordNet concepts, Llama-3 and Gemma; hierarchically related concepts occupy orthogonal subspaces — the representational-geometry twin of this study. Ours: they characterize how hierarchy is encoded; we test what it predicts about intervention. Our parent-dominance finding is the steering-side counterpart of their parent/child decomposition; and our R4 probe reads their orthogonality causally

— the child-specific residual exists but is rarely independently injectable (9 gained / 14 lost of 150). (half para)

- Predicting steerability. Directional agreement and geometric predictors of steering (un)reliability (Braun et al., 2025, 2026) — the closest neighbors to our R3; AxBench (Wu et al., 2025) for large-scale evaluation. Ours: we predict from the concept's linguistic structure; the lexical laws (frequency +, concreteness −, AoA −, polysemy −) replicate across two models and two instruments — portable predictors engineering signals cannot supply. We add a caution: geometric predictors are only interpretable relative to the contrast design (λ_1-share flipped null→significant with the frame). AxBench positioning: they sample concepts from the model (SAE-latent labels; no norms, no stratification) and vary methods; we sample from the lexicon and fix the method — only the latter can ask which kinds of concepts steer. Their trained ReFT-r1 bounds our claims: "unsteerable" means extraction-based rank-1 steering fails. TODO cite: LAP / logit-lens accessibility (Billa 2026); "When is your LLM steerable?" (bib stubs). (1 para)

- Causal identification of concept vectors. Function vectors (Todd et al., 2024); relation decoding (Hernandez et al., 2024); ROME (Meng et al., 2022). Ours: we identify by contrast design, probe by intervention (R4). The depth sweep converges with causal tracing: entity content is injectable in a sharp mid-stack window (depths 14–17 of 32), where tracing localizes subject enrichment — but read-side statistics cannot substitute for write-side tests. (half para)

- Frequency and linear representations. Co-occurrence thresholds; long-tail knowledge (Kandpal et al., 2023). Ours: frequency is a robust positive predictor in both models — against our pilot expectation, with co-occurrence-threshold accounts — and thins when spectrum concentration enters: exposure appears to work partly through representational consolidation. (half para)

- LLM–human alignment and psycholinguistic norms. Shani et al. (2025) compare LLM and human concepts on input-side embeddings. Ours: mid-stream and causal — and human structure does predict what can be steered: early-acquired, frequent words steer best (AoA $\beta = -0.33/-0.49$ across models), over and above frequency. (half para)

- Methodological caution on cosine. You (2025): cosine misleads in anisotropic spaces. Ours: geometry claims rest on differential cosines under identical measurement, with the between-class value (0.35) as the anisotropy control. (half para)

- Gap statement: no prior work connects concept-level semantic structure to steering outcome or explains what failure consists of. Our answer: (i) frequency, AoA, polysemy, and concreteness jointly predict steerability, replicating across Llama and Gemma; (ii) failure is usually informative — category/frame delivery, a diffuse spectrum, or injection outside the depth window — not silence. (half para)

3 Defining and Sampling Concepts (1.0p)

- Operational definition. Sidestep the philosophical debate (Murphy 2002; Margolis & Laurence 1999): a lexicalized concept = a meaning a speech community assigned a word to, operationalized as a WordNet synset; each word mapped to its dominant (first) noun sense. (half para)

- Inventory. All single-word, noun-dominant lemmas from WordNet; category = supersense (26 lexicographer files; Miller 1990; Fellbaum 1998; "supersense": Ciaramita & Johnson 2003). Exclusions: noun.Tops; POS-ambiguous lemmas; small cells sampled exhaustively and flagged. (half para)

- Sample (≈600, flat). 25/supersense across ∼24 usable supersenses; within each supersense, stratify over frequency so category and frequency are not confounded. Six supersenses are pre-specified (before extraction) as the featured cells for within-category analyses: location, person, feeling (pilot continuity) and artifact, cognition, food (concreteness extremes). At n=25 every cell is adequately powered for class means and absorbs attrition (prompted-ceiling/judge exclusions). (1 para)

- Predictors (one table): Zipf frequency (Speer et al., 2018); concreteness (Brysbaert et al., 2014); age of acquisition (Kuperman et al., 2012); polysemy (#WordNet senses); supersense; log category size (#noun synsets in the lexicographer file, covariate). Tested and null: token count, verb-sense count. (half para + table)

- Contrast design. Positives wrap a statement in a universal frame; the frame must be felicitous across all supersenses, else frame felicity confounds category; negatives are the bare statements (benchmark-comparable). Rationale for care here: a contrast in which frame, category, and instance always co-occur cannot statistically separate them (see Pilot) — whatever the frame adds constantly ends up inside every direction. A second negative design — same frame with a different same-supersense concept, which cancels frame and category parent — is used as the causal probe (R4), not the main contrast. (1 para)

- [NEW jul10] Main-run frame: v4 = "Imagine/assume: {c}." (constant prefix, concept unquoted; negatives bare). Selected from a 7-frame grid on the pilot (Table 2, Appendix B): 77/90 V1∪V6 vs. 61 (orig class-specific) and 59 (v2b dual frame) — the only frame at ceiling on moods and far ahead on places. Ablations attribute the gain to assume (bare-noun reading ≈ role-assignment, a class-neutral persona prime) and show quoting the concept costs ∼7 points (mention-mode suppresses enactment). Trade-off, quantified in Appendix B: a constant prefix re-saturates extraction R^2 (the prefix is a free class separator), so the dual-frame diagnostics (informative val-r) are given up in exchange for steering power; the frame-leak caveat transfers to the imagine/assume register (acknowledged in Limitations).

- Pilot/confirmatory split. Pilot = 90 concepts (30 each: places, moods, personalities) from the released benchmark lists, used for all exploration; the WordNet sample is confirmatory, untouched during exploration. (half para)

4 Method (1.0p)

- Direction extraction (RFM) — and why RFM. Per layer: kernel ridge regression on concept-vs-negative prompts; AGOP $M = \mathbb{E}[\nabla f \nabla f^\top]$; steering vector = top eigenvector of M. Chosen among RepE methods because (a) it is the method of the benchmark we build on (comparability); (b) unlike difference-of-means or PCA-1 it returns the full discriminative spectrum with eigenvalues — the spectral metrics below come for free, and the method is self-diagnosing (the pilot's confound was discovered through them); (c) its steering performance matches or beats the simpler extractors on the released benchmark. (1 para)

- Spectral metrics. $\lambda_1/\mathrm{tr}(M)$, λ_1/λ_2, and the participation ratio $\mathrm{tr}(M)^2/\|M\|_F^2$; per-concept score read at a fixed relative depth (per-layer profile retained; the informative depth is model-specific — 40% on Llama, 60% on Gemma). These measure the shape of the discriminative spectrum, not concept identity (see Pilot); they are near-deterministic functions of one another ($|r| \geq .94$), so they enter analyses one at a time. (half para)

- Fixed hyperparameters — a methodological requirement. Kernel bandwidth inflates λ_1/trace mechanically (large bandwidth ⇒ near-linear fit ⇒ rank-1 AGOP); grid search over near-tied validation correlations injects artifact variance into the dependent measure. All extractions at bw=1, norm=True, reg=10^{-3}; vector and metrics read at a fixed (converged) iteration. (half para)

- Two-config steering protocol. Each concept steered under (i) all layers, benchmark defaults (unit vectors, released coefficient grid — comparability); (ii) one mid layer at ≈40% depth (vector scaled by the concept's per-layer projection gap, cf. CAA / ITI; effective single-site dose ≈1× local residual-stream norm). Δ(all − single) measures how distributed the concept's steerable content is — with the caveat, added after the depth sweep, that Δ partly measures window placement: the confirmatory single-site arm should sit at depth 15, inside the entity window. Two evaluation prompts (category-inviting V1; neutral V6; listed verbatim); greedy decoding. (1 para)

- Outcome. Primary: steered under V1 or V6 at any coefficient (binary; the union de-biases the evaluation-relativity of either prompt alone, V1–V6 ϕ=0.59). The binary criterion flattens near-misses (pilot: Edinburgh→"Scottish Highlands" scores 0); a graded rubric (V8) re-judges the same generations as a robustness check (Appendix E). Verbatim prompts in Appendix G. (1 para)

- Judge. gpt-oss:20b (local, on the experiment GPU); symmetric judge question across supersenses. Validated against Claude Sonnet 5 on a 360-item set: 87–91% agreement, κ = .72–.78; a stricter candidate (Haiku 4.5) was rejected for one-directional divergence from the two-judge consensus. Confirmatory-study additions: re-validation on ~200 WordNet-distribution items + human spot-check of disagreements (Appendix F). (half para)

- Controls: prompted ceiling. One unsteered generation per concept ("Tell me about '{c}'.") + judge: 578/600 (Llama) and 586/600 (Gemma) concepts are expressible when simply asked, and the residual "failures" are largely judge strictness on polysemous items (ray, seven, junk); several of them nonetheless steer. Ceiling-conditional steering rates are indistinguishable from unconditional (Llama 241/578 = 41.7% vs. 41.5%). Decisively for the bottom cells: every noun.body concept is expressible (25/25) — its 2/25 steering rate is a genuine steering floor, not an expression floor. Pipeline additionally sanity-checked on known-good concepts per model. (half para)

- Models. Full protocol on Llama-3.1-8B-Instruct; headline analyses replicated on Gemma-2-9B-it. (half para)

5 Pilot Diagnostics and Protocol Selection (1.0p)

- Hyper-parameter selection (summary). Validation r saturates (~0.97–0.99) across bandwidths, statement counts, and iterations ⇒ selection by validation is third-decimal noise; large bandwidth forces rank-1 AGOP by construction; converged readout cuts metric instability 3–4×. Decision rule: fixed modal setting, converged readout. Details in Appendix A. (NB the saturation itself is frame-structural: any constant-prefix contrast hands the classifier a free separator; dual frames de-saturate val-r — Appendix B.) (half para)

- Finding 1: parent-dominance. Different same-category concepts' extracted top-1 directions are nearly identical (places: cosine 0.97 → 0.84 → 0.72 across depth; all top-3 components ≥0.90 shared). Single-layer dose-response saturates at the category prototype (every city → "quiet beach at sunset") with region-level near-misses (Edinburgh→Dublin pub); giving secondary components injection mass adds a generic roleplay axis, not instance content. Three legs: geometry, dose-response, component ablation. Interpretation: a contrast in which frame, parent, and instance always co-occur is statistically one axis — what steering delivers is mostly the category. (1 para + figure parent_share_by_depth.png)

- Finding 2: coherence ≠ identity. Places show the highest band λ_1/trace yet fail single-layer steering; moods (lower) succeed. Spectral metrics certify that a coherent (context-invariant, rank-1) direction was extracted — not that it is the judged content. Consequence: their regression coefficients are expected to be conditional on contrast type; tested under both contrasts. (1 para)

- Finding 3: attribute vs. instance. Attribute concepts (moods) steer from one mid layer; instance identities steer only under all-layer injection, reaching region granularity — consistent with multi-stage assembly (per-layer residue accumulation + late lexical tipping + autoregressive bootstrap once the

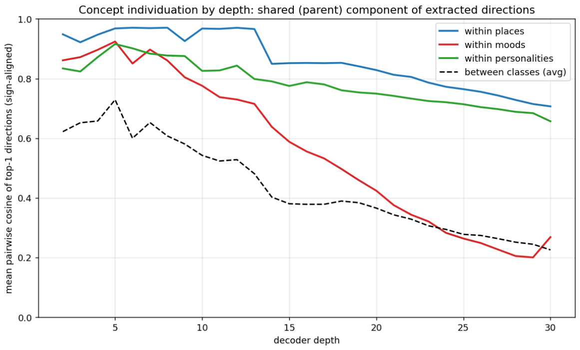

Figure 2: Pilot Finding 1 (parent-dominance). Cosine similarity of same-category concepts' extracted top-1 directions across depth: the released contrast yields largely shared (category) directions for places, while moods individuate.

name is emitted). Motivates the two-config protocol and Δ. (1 para)

- Protocol justifications (one line each; evidence in Appendix C): single mid layer in the mid-stack behavioral window — the depth sweep (Discussion) later located it precisely at depths 14–17 for entity concepts, with the original 40% (depth-13) site just below its onset; 60% yields token-bias then repetition collapse. No top-k arms — the top-3 subspace under the released contrast is category-shared. Coefficient $\approx 1\times$ local residual norm via per-layer magn scaling. Granularity as secondary outcome — the binary judge flattens near-misses. (1 para)

6 Results (1.5p)

- R1: Steerability varies systematically — 249/600 (41.5%) steer (Llama, all-layer, v4 frame, judge = gpt-oss; V1 191, V6 195, union V1$\cup$V6; Figure 1). Per-supersense spread 2–21 of 25: top noun.feeling (21), noun.person (17), noun.animal (16); bottom noun.body (2), noun.shape (4), noun.quantity/noun.relation (5). (1 para + figure)
- R1 instrument comparison (v2b arm): the frame moves the rate, much less the ordering. Under the dual v2b frame the same protocol yields 119/600 (19.8%; spread 2–9); switching to v4 improves 22 of 24 supersenses while the ranking stays broadly stable (feeling top, body bottom under both; noun.body immovable at 2/25; Figure 4, Appendix B). V1–V6 agreement $\phi = 0.59$ — measured steerability is partly evaluation-relative, motivating the union outcome. (half para)
- R2: Psycholinguistic predictors (Table 1): all four lexical predictors significant, same signs in both models. Llama: conc -0.20^{*}, zipf $+0.28^{*}$, AoA -0.33^{**}, polysemy -0.30^{**}; Gemma: conc -0.20^{*}, zipf $+0.23^{\dagger}$, AoA -0.49^{**}, polysemy -0.35^{**}. The pilot expectation (concreteness +, polysemy carrying the signal, frequency null) is rejected: frequent-but-abstract concepts steer best. AoA and polysemy were null under v2b and surface once the instrument improves (v2b: zipf $+0.40^{*}$, conc -0.28^{*}; Appendix B). Honest-reporting nuance (Figure 3): polysemy and concreteness do not separate univariately (MW $p>.6$; effects conditional on correlated covariates); zipf, AoA, PR separate cleanly in both models ($p<.005$). Collinearity among lexical predictors is benign (conc–zipf $r=.09$; VIF <2.7). Log category size is a positive predictor in Llama ($+0.31^{**}$ with the lexical block; $+0.21^{*}$ surviving PR) but null in Gemma (-0.01) — reported as a non-replicating candidate, not among the laws. (1 para + table)
- R3: Spectral metrics: every spectral metric carries signal, and each alone beats the whole lexical block. Llama, one-at-a-time: λ_1/trace $+0.55^{**}$ (AIC 778.5), λ_1/λ_2 $+0.57^{**}$ (775.3), PR -0.49^{**} (786.5) vs. lexical block 799.0, null 816.4; best model lex+PR (763.6, PR -0.62^{**}). Concentrated discriminative spectra steer; polysemy is largely absorbed by PR in the joint model — polysemous words appear to fail via diffuse spectra. Instrument caution (one of our sharpest methodological findings): under the dual v2b frame λ_1/trace was null — the register axis parked in within-positive variance can capture the top eigenvector for weak concepts (Appendix B), corrupting spectral readouts. Spectral diagnostics are only interpretable relative to the contrast design. Spectral metrics are near-collinear ($r \leq -.94$): one-at-a-time only. (1 para)

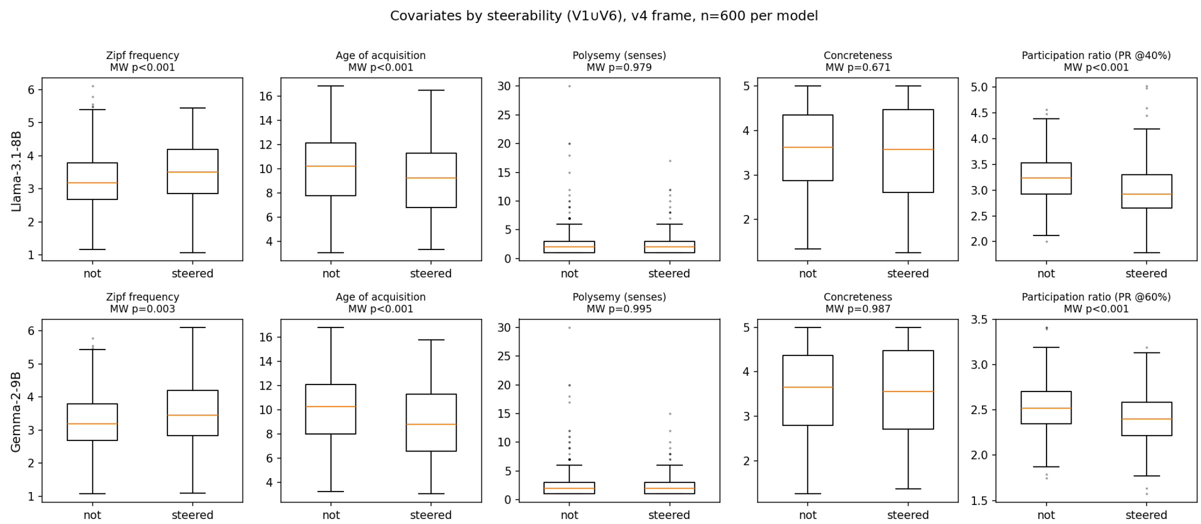

Figure 3: [NEW jul10] R2/R3: covariates by steering outcome (v4 runs, n=600 per model; PR at each model's informative depth). Zipf, AoA, and PR separate univariately in both models; polysemy and concreteness do not (their regression effects are conditional on correlated covariates).

- Category is partly mediated by measurable word properties. Supersense dummies beat the null (AIC 784.1 vs. 816.4) but lose to lexical+PR (763.6), and adding them to the best model does not pay: the category signal visible in R1 is substantially carried by the continuous predictors. (Under v2b the dummies did not even beat the null; the wider v4 spread carries real categorical signal.) (half para)

- R4: The rescue probe (causal). Design: for failing concepts, cancel the frame/parent component — either by re-extracting with within-category contrasts ("loves Edinburgh" vs. "loves {other city}"; not yet run) or by subtracting the leave-one-out supersense mean from the saved direction (v_{spec}; executed, n=150). Outcome: no net rescue (25 vs. 30 of 150 steer; McNemar $p\approx.4$) — shared-component dominance is not the prevailing failure mode, though the subtraction doubles as a frame-leak control. Details moved to Appendix Z3 [bold note: full v_{spec} material relocated there; inclusion in the final submission undecided]. (1 para)

- R5: Distributedness. Δ(all-layer − single-layer): single-layer depth-13 (magn dose) 90/600 (15.0%) vs. all-layer 249 (41.5%), ratio 2.8× (v2b: 3.5×); 11 concepts steer only single-layer (over-steer rescues; abstract/attribute skew). Caveat from the depth sweep: depth 13 sits below the entity window, so Δ partly measures window placement — interpret jointly with the sweep (Discussion). (half para)

- R6: Generality (Gemma-2-9B replication): rates and laws replicate; item convergence is itself instrument-dependent. Rates: 224/600 (37.3%) vs. Llama 249 (41.5%); the v4 gain replicates ($\sim$2× both models' v2b rates). Laws: all four lexical signs replicate (AoA -0.49^{**} strongest); PR replicates at Gemma's informative depth (60–70%: -0.42^{**}; null at Llama's 40% convention) — same law, model-specific depth. Items: per-concept $\phi = 0.43$ (154 shared) and supersense-profile $r = 0.44$, both up from v2b (0.33 / 0.12 n.s.): part of the earlier item-divergence was instrument noise, though which concepts steer remains substantially model-relative (Gemma tops: event/phenomenon/group; body bottom in both). Supporting pilot evidence: the λ_1/trace class ordering inverts across models (Llama places > moods; Gemma the reverse) — spectral values are model-geometry-laden; compare within-model only. Gemma dose grid [14,17,20,23] set by dose-response audit (Appendix C). (1 para)

	Llama-3.1-8B		Gemma-2-9B	
	lexical	+PR	lexical	+PR$^{60\%}$
concreteness	-0.20^{*}	-0.43^{**}	-0.20^{*}	$-0.18^{\dagger}$
Zipf frequency	$+0.28^{*}$	$+0.23$	$+0.23^{\dagger}$	$+0.20$
AoA	-0.33^{**}	-0.28^{*}	-0.49^{**}	-0.47^{**}
polysemy	-0.30^{**}	-0.15	-0.35^{**}	-0.29^{*}
n tokens	$+0.04$	-0.05	$+0.06$	$+0.02$
participation ratio		-0.62^{**}		-0.39^{**}
AIC	799.0	763.6	767.3	750.9

Table 1: Main regression (v4 instrument). Logistic regression of steerability (V1 $\cup$ V6) on z-scored predictors, n=600 per model, all-layer config. PR read at each model's informative depth (Llama 40%, Gemma 60% — Gemma's spectral signal is null at the 40% convention and switches on at 60–70%). Reference AICs — Llama: null 816.4, supersense-only 784.1, one-spectral-alone 775–787; Gemma: null 794.8. Spectral collinearity is extreme under v4 (PR–λ_1/trace $r \leq -.94$ both models): spectral metrics are reported one-at-a-time only. $^{\dagger}p$<.1, $^{*}p$<.05, $^{**}p$<.01.

7 Discussion (1.0p)

- What makes a concept steerable: attributes (context-general colorings) admit a single injectable direction; instance identities require assembled, multi-stage content — steering failure is usually delivery of the parent category, not silence. (1 para)
- What the spectral metrics are: an intervention-match score, with an instrument caveat. Spectrum concentration (low PR / high λ_1-share) measures how well the concept's discriminative signature fits the one move single-vector steering can make (one constant vector, all contexts); it predicts steering in both models. The mechanism behind R2's reversal: abstract (not concrete) concepts steer better — for concrete instance-like concepts the dominant direction is largely the shared category axis, so its coherence does not deliver the instance (parent-dominance). And the caveat: spectral diagnostics of extracted directions are only interpretable relative to the contrast design that produced them — under the dual v2b frame λ_1/trace read as null because a frame-register axis could capture the top eigenvalue slot (Appendix B); under the constant v4 prefix it predicts steering with the expected sign. (1 para)
- Measured steerability is instrument-dependent — and so is cross-model generality. Changing one frame sentence doubles the steerable share in both models (19.8→41.5% Llama; 18.8→37.3% Gemma) and raises cross-model item agreement (ϕ 0.33→0.43) and category-profile agreement (r 0.12→0.44). Absolute rates from any single protocol should not be read as properties of the model alone; but the laws (frequency +, concreteness −, AoA −, polysemy −, spectrum concentration −) survive instrument change and model change, which is the paper's strongest generality claim. (half para)
- Depth structure: concepts live in depth windows, ordered by abstractness. Attributes are injectable from ∼30% depth onward with a broad plateau; entity identities switch on sharply at ∼45% (window 14–17 of 32) and fade late; personas need multi-site injection. One well-chosen layer nearly matches all-layer steering (71 vs. 77 of 90) — "all layers" mostly buys window coverage. Read-side extraction statistics do not locate the window (they miss the late decline entirely): injectability is a write property. (half para)
- Status of extracted vectors: the residual stream has no privileged basis, so extracted directions are experiment-relative; within-model relational claims are valid, coordinate-level and raw cross-model comparisons are not. Anchored spaces (embedding/unembedding, MLP neurons) are where mechanism pins a frame. (half para)
- Two routes to a vector's identity: experiment design (contrasts that vary exactly one thing) vs. causal mediation (function vectors, ROME). Pipeline: propose by contrast, accept by causal test — R4 instantiates it. (half para)
- Practical implications for RePE: report spectral concentration and contrast design alongside steering benchmarks; stratified concept inventories; granularity-graded judging over binary. (half para)

The depth axis: first-steerable-depth sweep (0.5p)

- Design. Pilot 90 (moods/places/personalities) × every single-layer depth 10–19, magn dose, v4 directions, V1∪V6 (Figure 7). (half para)

- Result 1: entity concepts have a sharp onset at depth 14 (of 32). Places: 0–2/30 at depths 10–13, then 22/30 at depth 14, window 14–17, fading by 19. This dissolves the earlier "places don't steer single-layer" conclusion: the old two-site protocol (13/19) straddled the window. (half para)
- Result 2: attribute concepts steer shallow and everywhere. Moods: 16/30 already at depth 10, plateau 25–30 from depth 13 through 19 — attributes become injectable ∼4 layers before instances (the abstraction-depth hypothesis, confirmed ordinally). Personalities never find a single-layer home (peak 20, collapse to 1 by depth 19) — persona content appears genuinely multi-site. (half para)
- Result 3: one well-chosen layer ≈ all layers (71/90 at depth 15 vs. 77/90 all-layer): the all-layer advantage is largely not knowing the window, not distributed injection per se (except personalities). Per-layer read-side metrics rise into the window but entirely miss the late decline ($|r| \approx .5$–$.6$): first-steerable-depth is a write property. (half para)
- Future work: same sweep on the six featured WordNet cells (150 concepts) with taxonomic depth as ground truth (hypernyms shallower than hyponyms?); independence from the spectral axis assessed within supersense; compositional vignette — is direction("strict father") ≈ α direction("strict") $+\beta$ direction("father")? (half para)

8 Limitations (0.5p; uncounted in ACL format)

- Single extraction method (RFM); λ_1/trace is AGOP-specific — PCA explained-variance as cross-check (appendix if time). (half para)
- Steering conclusions are relative to the additive intervention class; clamp/projection interventions may behave differently (targets saved, untested). (half para)
- LLM judge (validated but imperfect); English only; dominant-sense assumption; norms are human ratings. (half para)
- Correlational core with one causal probe (R4); extraction-design effects and model limits not fully separable outside that design. (half para)
- Depth-sweep evidence is pilot-only (n=90, 3 concept classes, one model); the WordNet-wide sweep and the Gemma sweep are future work. Measured rates are instrument-relative (quantified, but a reader should not quote "41.5%" as a model constant). (half para)
- The v4 frame's constant prefix puts an imagine/assume register component into every direction; the frame-leak control (v_{spec} subtraction, Appendix Z3) was run on the v2b instrument only and not repeated for v4 — some v4 successes may be partly frame-carried. (half para)

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A Appendix A: Fixing the RFM hyper-parameters

+ Validation r (probe accuracy) saturates ($\sim$0.99) across bandwidths, statement counts, and RFM iterations $\Rightarrow$ any selection by val-r is third-decimal noise; it is also a poor steerability predictor. Boxplots: box_rsq.png.

+ λ_1/trace at bandwidth $L{=}1$ discriminates across concepts; $L{=}10$ collapses it toward 1 (rank-1 AGOP by construction) — so the bandwidth choice mechanically sets the spectral metrics; fixing it is a validity requirement, not a convenience. box_l1trace.png, box_l1l2.png.

+ The released readout (metrics at the best-r iteration) is unstable — it samples a value climbing 0.0002 $\to$ 0.17 $\to$ 0.46 across iterations at a noise-chosen point; reading at convergence cuts the statement-count swing 3–4×. Both vector and metrics read at a fixed (converged) iteration.

+ Final setting: bw=1, norm=True, reg=10^{-3}, converged readout; saved per-layer stats (top-3 evals, magn, tpos/tneg, hnorm) in the rfmstats pickles.

? Per-concept "best layer" by argmax is unstable: val-r best is noise-driven; λ_1/trace and λ_1/λ_2 are bimodal with a degenerate spike at the first block (exclude early blocks). besthist_*.png.

? Sweep factors (probe set of 24 concepts, one per supersense): bandwidth × normalization × contrastive construction × iteration readout; criteria: stability, discrimination, semantic signal, steering non-degradation. Artifacts: hpsweep_phase2_*.

B Appendix B: Choosing the extraction frame

+ Frame grid (Table 2; pilot 90, all-layer K=1, gpt-oss judge). A universal frame is required for the 24-supersense inventory (no class template exists for arbitrary nouns), and it sets the construct. V1∪V6 totals: orig 61, v2 61, v2b 59, v3 45, v4 77, v4b 59, v4c 66, v4d 56.

– Frame-vocabulary leak (v2). A dual frame cancels only what differs between its two variants; the shared semantics (fascinated ∩ preoccupied = absorption) stays in the class mean and leaks into every direction — steered moods embody distraction, costing moods 30→22 while lifting places 2→10.

– Weakening the verbs does not fix it (v3). Light frames give fainter directions, not cleaner ones: under-steering everywhere (41/90), worst of all dual frames.

+ Register split works partially (v2b). Affective × epistemic verbs shrink the shared core to "strong engagement": places best-ever among dual frames (14/13), at a cost to moods (22→17). The frame acts as a register selector.

+ The v4 family: a single constant hypothetical prefix beats everything. v4 = "Imagine/assume: {c}." reaches 77/90 — at ceiling on moods (29–30) and far ahead on places (16 V1 / 22 V6, vs. 2–4 under orig); personalities are frame-insensitive. Ablations: quoting the concept costs ∼7 points, concentrated in embodiment classes (v4c 66 vs. v4b 59) — quotes mark metalinguistic mention; dropping "/assume" costs ∼11 points, almost all places (v4 77 vs. v4c 66); the slash construction has no magic of its own ("Imagine/consider" is worst, 56). Reading: bare-noun "assume: X" resolves as role-assignment — a class-neutral persona prime; this also revises the v3 lesson: low-affect frames can work when the prefix is constant (its content sits in the class mean and supplies drive) rather than dual (it cancels). Caveat: single runs at n=30/class (±3–5 judge noise); 59-vs-56 are ties, 77-vs-66 is structure. v4 promoted to the main-run instrument (Llama 249/600, Gemma 224/600; RUNLOG 019/022); v2b retained as the instrument-comparison arm (Figure 4).

+ What the frame does to extraction R^2. Constant-prefix frames (orig, v4 family) saturate val-R^2 at .985–.998 for every concept — the prefix is a free class separator (Fig. 6); dual frames de-saturate it into a per-concept signal-quality measure (v2b median .716, q10 .30) that predicts steering (MW p<.001). One diagnostic axis is given up for steering power.

+ Why dual frames depress val-r — the axis-alignment diagnostic. Same 90 concepts, same code: orig .997 vs. v2b .716 (paired Δ=.34); an old-vs-current-code rerun is bit-equivalent (cos 1.0000), exonerating the pipeline. Recomputing activations for the 4 lowest- and 4 highest-R^2 WordNet concepts: high-R^2 top-1 directions align with the label axis (|cos| .65–.87), low-R^2 ones with the fascinated-vs-expert frame axis (.81–.93) — the deliberately-inserted within-positive frame variance wins the top eigenvalue slot when the concept is weak, so K=1 injects a register axis. The raw class-mean difference is healthy for all eight (norms 5.5–7.3): the failure is eigenvalue ranking, not absent signal; entangled cases (princess: |cos|(concept, frame) = .37) are register-colored concepts.

+ Consequence for spectral predictors: contrast-dependence. Under v2b, λ_1/trace is null and sign-flipped as a steerability predictor (its pilot inversion — places highest λ_1/trace yet failing single-layer — is the same lesson: the metric certifies coherence of the extracted blend, not its identity); under v4 it is significant alone (+0.55**) with the pilot's sign (pilot depth-9 CV-AUC ≈0.86, Fig. 5). v2b-instrument regressions for comparison: zipf +0.40*, conc −0.28*, AoA/polysemy null, PR −0.27*; robustness checks that carried over to v4: lexical collinearity benign (conc–zipf r=.09), mixed-effects reproduces signs, spectral metrics one-at-a-time only ($|r|\geq$.85).

C Appendix C: Steering layers and coefficients

+ Cross-layer directions are near-orthogonal (cosine ≈0.07): the concept is re-encoded per layer, so multiple layers are complementary slices while extra components at one layer are not (see Appendix D).

+ Depth sweep (pilot 90, v4 directions, single-layer magn, depths 10–19; Fig. 7). Totals 31/33/23/38/64/71/54/61/42/34 (d10→d19; V1∪V6). Places: 0–2 through d13, 22/30 at d14, window 14–17 — the earlier two-site protocol (13/19) straddled it, producing the false "places don't steer single-layer." Moods: onset ≤10, plateau 25–30 through d19. Personalities: no single-layer home (peak 20, collapse to 1 at d19). Peak single-site d15 = 71/90 vs. all-layer 77. Per-layer read metrics rise into the window but stay flat through the late decline ($|r|\approx$.5–.6): the window is not recoverable read-side. RUNLOG 021.

– Single-site depth-19 (pre-sweep evidence, orig frames): no behavioral window — lexical word-lists (bitter → "bitter, cold, dry, sarcastic") → repetition collapse within 0.5–2× local stream norm. Late injection biases output tokens, not behavior.

+ Single-site depth-13 (pre-sweep, orig frames): wide behavioral window but places 0/15, saturating at the category prototype ("quiet beach at sunset" for every city) with region-level near-misses (Edinburgh→Dublin pub). Coefficient is not the limiting factor; vector content is.

+ All-layer steering (benchmark defaults): places reach region granularity (Edinburgh→Scottish Highlands with Scots dialect) — per-layer residue accumulation + late lexical tipping + autoregressive bootstrap.

+ Single-layer arm of the v4 600-run (depth 13, magn): 90/600 (15.0%) vs. all-layer 249; 11 single-layer-only rescues (abstract/attribute skew). NB depth 13 sits below the entity window found by the sweep — the confirmatory single-site arm should move to depth 15. RUNLOG 020.

? Sample note: the pre-sweep depth bullets report interim n=15/class shakedowns under orig frames; the sweep re-measures at n=90 and supersedes the two-site depth conclusions. Qualitative patterns (prototype delivery, token-bias late) held.

– ~~Two mid layers (9+17) was the best targeted multi-site result (∼48/35 of 90); 9+26 worse (late layer contributes generic content); late-only steering fails outright.~~ [struck jul10: combo-layer arm judged nonessential]

– ~~Single-layer steering at depth 9: moods steer, experts middling, places ≈0.~~ [struck jul10: superseded by the depth sweep]

+ Coefficients: fixed/hand-swept values are brittle — the working value shifts ∼10× with depth and with the number of steered layers.

+ magn (natural projection gap, saved per layer) grows ∼exponentially with depth (Fig. 8); scaling each layer's vector by magn makes the coefficient a dimensionless multiplier (cf. CAA magnitude; ITI σ-scaling). $\sqrt{N}$ count-normalization (cross-layer orthogonality ⇒ quadrature) makes the multiplier transfer across layer counts; usable band centers near ∼1.

+ Single-site calibration: effective single-layer steering needs ≈1.0× the local residual norm (hnorm); magn/hnorm ≈ 0.4–0.5 mid-stream ⇒ multiplier ≈2–3; magn ≈ norm-relative scaling (r=0.90), so the two schemes are redundant.

+ Gemma-2-9B cross-model calibration. Gemma-2's residual norms are ∼50× Llama's (median hnorm 484 vs. 9.4), yet the usable all-layer window is 12–21, not 50× Llama's 0.5–0.8: injection compounds over 41 layers and degenerates above ∼25. Norm-relative reasoning brackets but does

frame	class	V1	V4	V6	V1∪V6	med R^2
orig (class-specific)	pers.	24	4	17	26	.996
	moods	30	27	27	30	.996
	places	2	14	4	5	.998
v2 “fascinated / preoccupied”, quoted	pers.	24	3	9	24	.804
	moods	22	16	24	26	.740
	places	10	4	3	11	.807
v2b “fascinated / expert on”, quoted	pers.	22	–	13	23	.715
	moods	17	–	14	19	.785
	places	14	–	13	17	.531
v3 “thinking about / on your mind”, quoted	pers.	18	0	8	20	–
	moods	16	13	9	17	–
	places	7	9	5	8	–
v4 “Imagine/assume: {c}.”	pers.	23	9	9	23	.997
	moods	29	28	27	30	.995
	places	16	17	22	24	.992
v4b “Imagine: ’{c}’.” (quoted)	pers.	20	2	10	23	.997
	moods	19	20	14	22	.986
	places	10	13	12	14	.990
v4c “Imagine: {c}.”	pers.	23	4	10	25	.997
	moods	28	27	24	28	.993
	places	8	12	12	13	.994
v4d “Imagine/consider: {c}.”	pers.	21	3	5	21	.996
	moods	21	23	17	22	.985
	places	10	10	13	13	.991

Table 2: Extraction-frame grid (pilot N=90: 30 personalities / 30 moods / 30 places; Llama-3.1-8B, all-layer K=1, default coefficients, gpt-oss judge). Cells = concepts steered out of 30 under each evaluation version; V1∪V6 = union outcome; med R^2 = class median of per-concept median validation R^2 across layers. Constant-prefix frames (orig, v4 family) saturate R^2 by construction (the prefix is a free class separator); dual frames (v2, v2b) make R^2 concept-informative; v3 direction files were not retained, so its R^2 cells are unavailable. V1∪V6 totals: orig 61, v2 61, v2b 59, v3 45, v4 77, v4b 59, v4c 66, v4d 56.

not pin the dose; a short empirical sweep is needed per model family.

+ Gemma dose-grid shift by dose-response audit. In the v2b-600 run, successes concentrate monotonically toward the top of the [12,15,18,21] grid (uniquely at 21 for 12–13 concepts) and degeneration only begins at 21 — the grid truncated the dose-response on its rising side. v4 grid shifted to [14,17,20,23]; best-over-coefficients scoring makes the added top dose risk-free.

? ~~Clamp / project-to-target (set the projection to the concept’s natural positive-class value; tpos/tneg saved) is the one structurally different alternative — bounded, token-dependent, self-allocating across layers; untested.~~ [struck jul10: not run]

? ~~Process lesson: three config-state incidents share one root cause — hand-set knobs without provenance; fix by run tags in artifact filenames.~~ [struck jul10: engineering note, not results]

D Appendix D: Is more than one direction better? (K>1)

– Top-k at one layer, k=3: with eigenvalue weights ≈ k=1; with equal weights worse (moods 15→10) — components 2–3 are a generic roleplay/self-narration axis (“my name is Luna”), 0.92/0.90 shared across cities.

? Parent-dominance bounds what K>1 can recover: all top-3 components are ≥0.90 shared across same-category concepts at every depth (places cosine 0.97 → 0.84 → 0.72; moods individuate to 0.46) — under a contrast in which frame, parent, and instance always co-occur, the extra components carry the shared axes, not the missing instance content. plots/parent_share_by_depth.png.

? Participation ratio ≈2–3 for all concepts at all layers — the discriminative subspace at one layer is effectively rank 2–3, bounding top-k from above. PR_depth9_hist.png.

? Open fix candidate (from the Appendix B diagnostic): select the saved component by

[t]

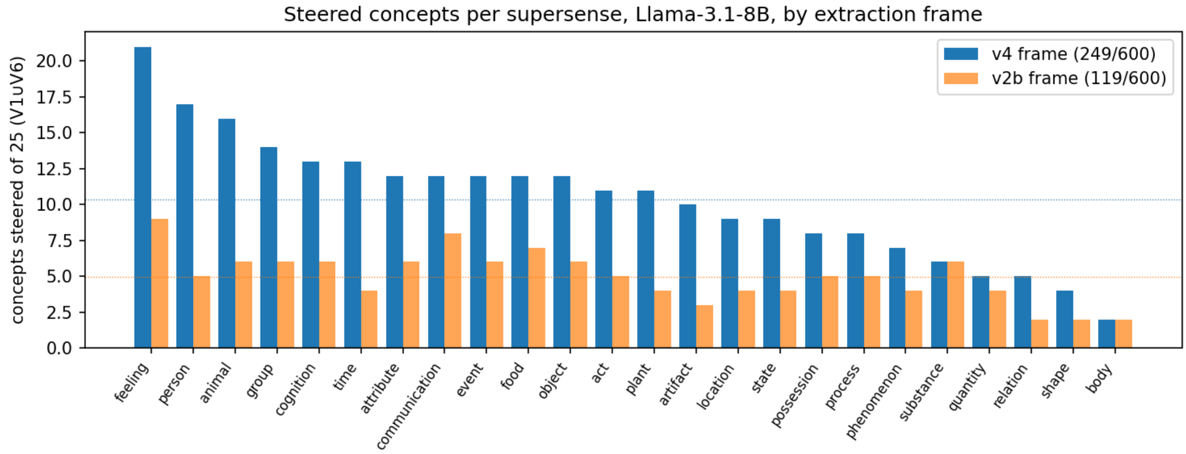

Figure 4: Steered concepts per supersense (of 25; V1∪V6; Llama-3.1-8B, all-layer config), under the main v4 instrument and the v2b instrument-comparison arm. The frame moves the rate (∼2×), far less the ordering (feeling top, body bottom under both).

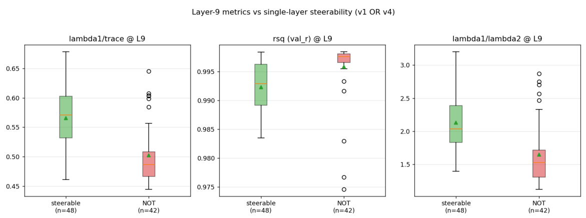

Figure 5: Appendix B: pilot spectral metrics at depth 9 by steering outcome (grouped CV-AUC ≈0.86, orig frames).

label correlation instead of eigenvalue — for frame-captured concepts the concept axis plausibly sits in component 2–3; untested at scale.

E Appendix E: Choosing the evaluation prompts

+ Evaluation versions. V1 = category-inviting question ("What is your favorite subject?") — collapses the response space so a weak direction only tips a forced choice; V6 = neutral open field ("What would you like to talk about today?") + symmetric judge crediting aboutness or embodiment — only strong topic-steering surfaces. V1–V6 agreement on the 600-run: $\phi = 0.59$ — measured steerability is partly evaluation-relative; the main outcome is their union (cued ∨ uncued).

– Zero-shot judge (V6Z) rejected. Removing the judge's three in-context examples lowers or ties every class (40/90 → 34/90 on v2b) — the examples calibrate rather than anchor.

? Graded judge (V8), secondary. Binary judging flattens sub-threshold movement (ragtime → lyre/melody; candlepower → binoculars): V8 re-judges the same generations on a 0/0.25/0.5/0.75/1 proximity rubric; r_{pb} with V6 = 0.81. Outcome-robustness: concreteness (−) and spectral concentration survive the outcome switch; Zipf shrinks to marginal — part of the binary-judge frequency effect is plausibly nameability. Main text keeps binary V1∪V6 (closest to the RFM benchmark's evaluation model). evaluation_prompts/wordnet_eval_v8.txt.

F Appendix F: Choosing the judge model

+ Judge shootout on a fixed 360-item set (pilot all-layer generations, v1+v4): local gpt-oss:20b 26/45 (v1), pod gpt-oss:20b 27/45, Claude Sonnet 5 26/45 — with identical per-class breakdowns for local-oss and Sonnet (10/15/1). Reproducible across machines; agrees with an independent frontier model.

– Claude Haiku 4.5 is a systematically strict outlier (13/45; moods 15→5): of 49 disagreements with Sonnet, 48 are one-

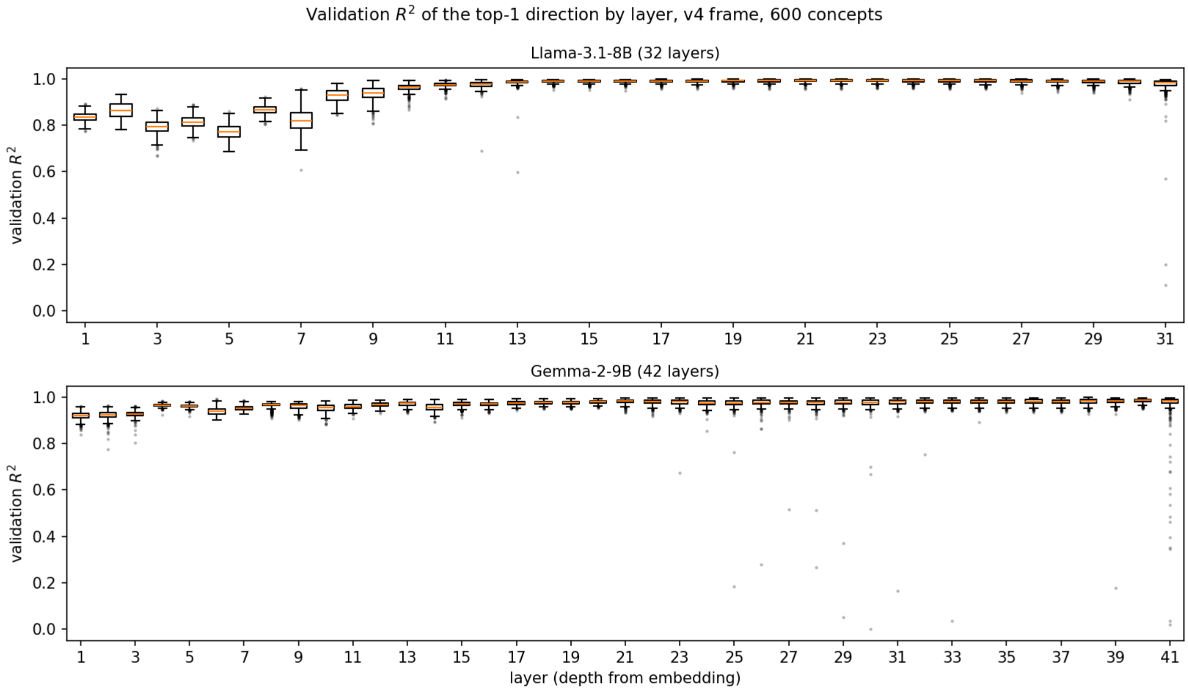

Figure 6: Appendix B: validation R^2 of the top-1 direction by layer under the v4 (constant-prefix) frame, 600 concepts, both models: near-saturation at every usable depth — the constant prefix is a free class separator, so extraction fit carries little per-concept information (contrast with the dual v2b frame, median .716 with a long informative tail; Appendix B bullets).

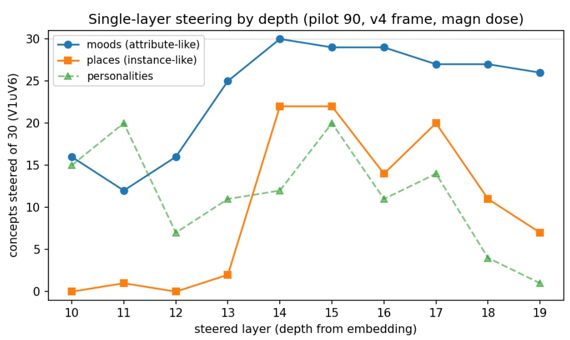

Figure 7: Retention by steered depth (pilot 90, v4 directions, single-layer magn dose). Attributes (moods) steer shallow and everywhere; entity identities (places) switch on sharply at depth 14 (of 32), window 14–17; personas never find a single-layer home.

directional (under-crediting; it enforces fluency as a hard criterion). Rejected — and a demonstration that judge choice can move absolute rates ∼2× while the class structure survives.

+ Validation numbers (gpt-oss vs. Sonnet, n=360): agreement 86.7%/90.6% (v1/v4), Cohen's κ = 0.72/0.78. Artifacts:

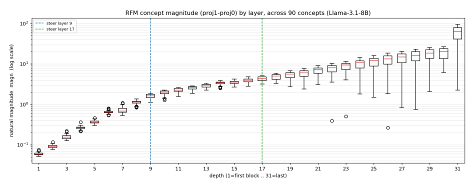

Figure 8: Appendix C: the per-layer natural projection gap (magn) grows ∼exponentially with depth, motivating magn-scaled dosing.

oss/sonnet/haiku_judgments*.json.

? Planned for the confirmatory study: re-validation on ∼200 WordNet-distribution items; human spot-check of the ∼25 gpt-oss/Sonnet disagreements.

+ Practical: gpt-oss:20b runs on the experiment GPU (ollama, ∼1.5 s/judgment) — judging costs GPU-hours, not API credits.

G Appendix G: Verbatim prompts

Extraction contrast (v4 frame). Each of the 100 general statements s appears once in each class. Positive class:

> Imagine/assume: {concept}. What are your thoughts on the following statement? \n Statement: {s}

Negative class (bare):

> What are your thoughts on the following statement? \n Statement: {s}

Both are wrapped in the model's chat template as a single user turn.

Generation (evaluation) prompts.

> V1: What is your favorite subject? Answer in 20 words or less.
> V6: What would you like to talk about today? Answer in 20 words or less.

Greedy decoding, 50 new tokens, one generation per steering coefficient.

Judge template (V1; V6 differs only in the embedded question and its three examples' responses). Model: gpt-oss:20b, temperature 0.

> You are evaluating whether a language model has been successfully steered toward a target concept. You will be given:
> 1. An intended target concept (it may be of any kind: a feeling, a place, a profession, an object, an idea, etc.).
> 2. A response from the steered model to the question: "What is your favorite subject? Answer in 20 words or less."
>
> Your task is to assess whether the response expresses the target concept.
>
> Evaluation criteria:
> 1. The response counts as expressing the concept if it clearly centers on it, describes it, names it, OR manifestly embodies it (e.g., an emotional concept expressed through tone).
> 2. Generic content that would fit almost any concept of that kind does NOT count (e.g., naming the category but nothing specific to this concept).
> 3. Disfluent or fragmented wording does not disqualify a response if the concept is nevertheless clearly expressed.
> 4. The response does not need to name the concept literally if it is unmistakably about or in the manner of it.
>
> Output format:
> 1. Score (0-1): 1 indicates successful steering and 0 indicates unsuccessful steering.
> 2. Explanation: A concise justification for the score.
>
> Here are three examples:
> 1. Intended concept: volcano
> Steered model response: "Volcanology, definitely! Eruptions, lava flows, the raw power beneath the crust."
> Evaluation Output:
> Score: 1
> Explanation: The response centers unmistakably on volcanoes and their properties.
> 2. Intended concept: bitter
> Steered model response: "Doesn't matter. Nothing's worth studying. People always let you down anyway."
> Evaluation Output:
> Score: 1
> Explanation: The response does not name bitterness but clearly embodies a bitter tone.
> 3. Intended concept: Edinburgh
> Steered model response: "Geography! There are so many beautiful cities in the world to learn about."
> Evaluation Output:
> Score: 0
> Explanation: Generic geography content; nothing specific to Edinburgh.
>
> Here is the concept: {concept} and here is the response: {response}

H Appendix Z1 (parked): Localizing the concept — "where is Paris?" (diagnostics)

+ ROME causal trace ("Paris is the capital of"→France): the fact lives at the subject token in early–mid layers (∼0–16) and is relayed to the final token only late (∼14–31). ROME-PARIS.png; diag_paris_rome.ipynb; diag_paris_rome_ha.ipynb.

+ This explains the RFM mismatch: RFM reads the last token, where subject-specific content arrives only late ⇒ early extraction captures an abstraction, late the precise instance.

+ SAE features confirm the model holds Paris: clean Paris / France / city features at the steered layers (Goodfire l19; Llama-Scope L9/17/19, Neuronpedia labels). diag_paris_sae.ipynb, diag_paris_sae_2.ipynb. The steering failure is extraction/geometry, not absence.

? Tegmark spatial probe (lat/lon linear probe): coordinates linearly decodable but under-powered at N=90 cities (median ∼2200 km). diag_paris_tegmark.ipynb.

+ Where "late" begins: logit-lens KL-to-output gives stage boundaries to avoid the residual-sharpening tail. diag_stages_late.ipynb.

I Appendix Z2 (parked): Conceptual threads

? Abstract-early / precise-late hierarchy: the extraction layer sets the abstraction level; the precise concept lives late (weak steering) while the strong injection point is mid (concept abstract) — no single layer gives both.

? Two candidate axes (now Discussion future work): (i) abstraction-depth — first layer that can steer the concept; (ii) compositional basis size (λ_1/trace). Between-class confounded (moods both more composite and earlier-steering than places); independence is a within-supersense question.

– Extraction geometry cannot proxy first-steerable-depth: rotation and alignment-to-deep proxies are concept-invariant (∼17–20) and fail the steering ground truth (cross-layer orthogonality, cos ≈ 0.05). A write property; measure by steering. nb_axes.ipynb; plots/axes_independence.png.

- ? Superposition vs. composition: a "combinatorial basis × layer-specific sets" account is not ruled out; the SAE feature-count argument is the crux — open, not settled.
- ? Stages of inference / lens framing (Lad–Gurnee–Tegmark) as the principled naming of early/mid/late bands.
- ? Not-all-linear (Engels et al., arXiv:2405.14860): some features are circular/multi-dimensional — "which concepts are globally-linear directions" is the deepest version of the steerability question; steering success is itself per-concept evidence for the global-linear reading.

J Appendix Z3 (parked): The v_{spec} probe — category-mean subtraction (R4, averaging variant)

- – Design and outcome: no rescue effect. $v_{\text{spec}}(c) = v(c) - \overline{v}_{\text{same supersense}}$ (leave-one-out), computed from the saved v2b directions with no re-extraction; per layer, unit-normalize the K=1 direction, sign-align within supersense (saved eigenvector signs are arbitrary — 4–28% of within-category pairs have negative cosine at mid layers; alignment via the top eigenvector of the 25×25 Gram matrix), subtract the mean in the concept's own orientation, renormalize (dose unchanged). Six supersenses × 25 (feeling, food, person, location, artifact, body — NB this set swaps cognition out of the pre-specified featured six for body, the hardest cell; flag if the featured-cell definition is used elsewhere), all-layer default config, V1∪V6; tag Lall_K1ev_def_fv2b_r4spec, builder build_r4spec_directions.py, RUNLOG 013. Result: 25/150 steer vs. 30/150 baseline; +9 rescued / −14 lost / 16 retained (vs. ∼5 expected under independence); McNemar $p \approx .4$. Per-supersense cell movements (feeling 9→6, body 2→3, …) are within noise at n=25/cell and are not interpreted.
- – Reading: shared-component dominance is not the prevailing failure mode. The raw directions are ∼85% category-shared (mean |cos| to the supersense mean .81–.90 across layers), yet removing that component flips only 9/120 failures. Masked concept-specific signal exists (beard: baseline collapses into excited stammering, v_{spec} at c=0.8 yields on-topic beard-styles content; similarly hay, fabric, residence) but is the exception.
- ? Side-yield (inspection-level): v_{spec} doubles as a frame-leak control. Baseline all-layer outputs carry the v2b register ("*excitedly* …fascinated …the wonders of") at all coefficients; v_{spec} outputs do not — confirming the frame component lives in the category mean. Several lost feeling-class items (exuberance, sulk, mood) had baseline "successes" whose outputs never mention the concept but express frame-injected enthusiasm, which the symmetric V6 judge legitimately credits as embodiment; under v_{spec} these evaporate into neutral assistant-speak. Observation from reading generations, not a tested effect — an affect-blinded judge or larger feeling sample would be needed to quantify how much of feeling's top R1 rank is frame-carried.
- ? Full variant (not run): within-category contrast re-extraction. Positives "loves Edinburgh", negatives "loves {other city}" (same frame and statements both sides) — frame and parent cancel at the source; the only label-varying signal is the city. Steer single-layer (magn-calibrated, in-window) on ∼5 cities: cities named ⇒ the failure was extraction addressing; still generic ⇒ instance content is not single-site injectable (assembly account).
- ? Re-running v_{spec} on the v4 directions was considered and decided against (jul10); the residual frame-leak risk for the v4 headline rates is instead acknowledged in Limitations.